

FINAL COMPLETE TECHNICAL PROJECT REPORT

AI-POWERED CAREER & EDUCATION GUIDANCE PLATFORM

Project Name: VisionPath	Architecture: Microservices + Spring Cloud Gateway
Backend Stack: Java 21, Spring Boot 3.5.5, Spring Data JPA	Frontend Stack: React 19, Vite 6, Tailwind CSS 4
Database: PostgreSQL (13 Dedicated DBs)	AI Engine: Google Gemini API (gemini-3.6-flash)

1. REPORT OBJECTIVE

Generate a complete final technical report explaining the entire VisionPath application from beginning to end. The report details the existing codebase structure, microservices, frontend pages, database schemas, security models, and Google Gemini AI orchestration without external additions.

2. PROJECT OVERVIEW

VisionPath is an AI-powered career and education guidance platform designed to assist secondary school and college students. The platform evaluates academic performance, aptitude, psychometric traits, interests, skills, career options, skill gaps, study requirements, resume quality, and career readiness.

Three Primary Roles:

- **Student:** Dual-tier access (School & College) to take 3 tier-specific assessments, manage skills, audit resumes, and execute study roadmaps.
- **Counselor:** Professional advisors who access assigned student dossiers and generate AI-synthesized progress summaries.
- **Admin:** System administrators who review counselor credentials, approve accounts, and manage platform metrics.

3. PROBLEM STATEMENT & 4. OBJECTIVES

VisionPath addresses subjective and unscientific career choices by providing objective multi-factor scoring, deterministic backend test checking, automated ATS resume parsing, and personalized 4-week study plan generation for identified skill deficits.

5. TECHNOLOGY STACK

Component	Actual Technology & Versions
Frontend	React 19.0.0, Vite 6.1.0, React Router DOM 7.1.5, Axios 1.7.9, Tailwind CSS 4.0.0, Lucide React 0.474.0, React Hot Toast 2.5.1
Backend Framework	Java 21, Spring Boot 3.5.5, Spring Cloud 2025.0.3, Spring Data JPA, Spring Security 6, JWT 0.12.6, Maven
Database	PostgreSQL (13 Dedicated Databases via Database-Per-Service pattern), HikariCP
AI / LLM Integration	Google Gemini API (gemini-3.6-flash / gemini-2.5-flash) via RestTemplate & JSON Sanitizer

6 & 7. COMPLETE BACKEND MICROSERVICE ARCHITECTURE

1. api-gateway (:8080) — com.visionpath.gateway

- Spring Cloud Gateway WebFlux reverse proxy, route dispatcher, global CORS, Actuator exposure.

2. auth-service (:8081) — com.visionpath.auth

- PostgreSQL: visionpath_auth_db. BCrypt password hashing, JWT creation/validation, password reset tokens.

3. user-service (:8082) — com.visionpath.user

- PostgreSQL: visionpath_user. User profile metadata, contact info, bio, avatar management.

4. student-service (:8083) — com.visionpath.student

- PostgreSQL: visionpath_student. Student academic profile, school/college fields, skill and interest records.

5. counselor-service (:8084) — com.visionpath.counselor

- PostgreSQL: visionpath_counselor. Counselor profiles, verification status, assigned students & appointments.

6. admin-service (:8085) — com.visionpath.admin

- PostgreSQL: visionpath_admin. Analytics dashboard, counselor approval/rejection, user account status toggles.

7. assessment-service (:8086) — com.visionpath.assessment

- PostgreSQL: visionpath_assessment. Question database, deterministic score calculation, grade assignment.

8. ai-service (:8087) — com.visionpath.ai

- PostgreSQL: visionpath_ai. Gemini client, 14 prompt builders, AI Mentor, question generation, study plan & resume audit.

9. career-service (:8088) — com.visionpath.career

- PostgreSQL: visionpath_career. Career library, required skill maps, career roadmaps, counselor booking.

10. skill-service (:8089) — com.visionpath.skill

- PostgreSQL: visionpath_skill. Skill catalog, student skill matrix, differential skill gap calculation.

11. study-plan-service (:8090) — com.visionpath.studyplan

- PostgreSQL: visionpath_study_plan. Multi-week study plans, weekly modules, granular task tracking.

12. resume-service (:8091) — com.visionpath.resume

- PostgreSQL: visionpath_resume. Resume metadata, ATS score records, extracted skills and feedback persistence.

13. notification-service (:8092) — com.visionpath.notification

- PostgreSQL: visionpath_notification. User alerts, status notifications, unread count tracking.

14. file-service (:8093) — com.visionpath.file

- PostgreSQL: visionpath_file. Document upload (resumes, ID proofs), storage metadata, binary retrieval.

8. API GATEWAY ROUTING MATRIX

Gateway Route	Target Service	Target URI	Purpose
/api/auth/**	auth-service	http://localhost:8081	Auth & Tokens
/api/users/**	user-service	http://localhost:8082	User Profiles
/api/students/**	student-service	http://localhost:8083	Student Profiles
/api/counselors/**	counselor-service	http://localhost:8084	Counselor Data
/api/admin/**	admin-service	http://localhost:8085	Admin Analytics
/api/assessments/**	assessment-service	http://localhost:8086	Assessments & Scoring
/api/ai/**	ai-service	http://localhost:8087	Gemini AI Engine
/api/careers/**	career-service	http://localhost:8088	Careers & Roadmaps
/api/skills/**	skill-service	http://localhost:8089	Skill Matrix
/api/study-plans/**	study-plan-service	http://localhost:8090	Study Roadmaps
/api/resumes/**	resume-service	http://localhost:8091	Resume Records
/api/notifications/*	notification-service	http://localhost:8092	Alerts & Notices
/api/files/**	file-service	http://localhost:8093	File Storage

9 - 15. FRONTEND ARCHITECTURE & USER ONBOARDING

Frontend Pages:

- *Student (8 Pages)*: Dashboard, StudentProfile, Skills, CareerAssessment, CareerRecommendation, CareerRoadmap, AIResumeAnalyzer, AICareerAssistant.
- *Counselor (4 Pages)*: Dashboard, AssignedStudents, StudentDetails, Profile.
- *Admin (3 Pages)*: Dashboard, CounselorMgmt, UserMgmt.
- *Auth & Public Pages*: SplashScreen, Login, Register, ForgotPassword, ResetPassword.

Registration & Security:

- Student registration captures Full Name, Email, Username, Password, Education Type (SCHOOL/COLLEGE), School Name/Grade or College/Degree/Branch/Year.
- Counselor registration captures Full Name, Email, Username, Password, Institution, Qualification, Experience, and LinkedIn Profile URL (Requires Admin Approval).
- Authentication utilizes BCrypt hashing and signed JWT tokens stored in localStorage. Axios interceptors attach tokens to HTTP requests.

16 - 20. ASSESSMENT & DETERMINISTIC SCORING ENGINE

School Student Suite (3 Assessments, 10 Questions Each):

1. *School Subject Knowledge*: Core Math, Science, English, Social Science, Computer Science.
2. *School Psychometric & Learning Attitude*: Persistence, time management, homework handling, group cooperation.
3. *Career Interest & Stream Exploration*: STEM, Medical, Commerce, Humanities stream mapping.

College Student Suite (3 Assessments, 10 Questions Each):

1. *Aptitude, Reasoning & Core Domain*: 3 Quantitative, 3 Logical Reasoning, 4 Core Technical Domain questions.
2. *Psychometric & Leadership Readiness*: Technical conflict resolution, code reviews, incident response, delegation.
3. *Career Specialization Analysis*: Full-Stack SaaS, Cloud DevOps, AI/ML Research, Cybersecurity specializations.

Scoring Decoupling: Correct answers are stored securely in the PostgreSQL `questions` table and hidden from student endpoints. When submitted, `AssessmentService.java` deterministically computes exact scores, percentages, and letter grades (A/B/C/D/F). Only afterwards does Gemini generate qualitative feedback.

21 - 31. AI FEATURES & ATS RESUME AUDITING

AI Features in ai-service (:8087):

1. *AI Mentor Chat*: Real-time conversational advisory via Gemini.
2. *AI Question Generator*: Dynamically generates 10 tier-tailored assessment questions.
3. *AI Result Analysis*: Qualitative strengths, weaknesses, and concept review feedback.
4. *Career Recommendation*: Multi-factor profile matching to industry career tracks.
5. *Skill Gap Evaluation*: Identifies missing technical competencies against target roles.
6. *Study Plan Generator*: Creates 4-week modular roadmaps with estimated task durations.
7. *ATS Resume Analyzer*: Scores resumes out of 100, extracts skills, and highlights lagging competencies.
8. *Counselor Student Summary*: Synthesizes student dossiers into executive counseling notes.

Automated Remediation Action: If the ATS Resume Analyzer identifies lagging skills, clicking '**Generate Study Plan for Lagging Skills**' automatically builds a 4-week remedial study roadmap saved in PostgreSQL and displayed on the student's Career Roadmap page.

32 - 37. COUNSELOR & ADMIN GOVERNANCE

- **Counselor Application Workflow:** Counselors register with institution details and LinkedIn credentials. Account status remains PENDING until reviewed and approved by an Administrator in `CounselorMgmt.jsx`.
- **Counselor Dossier Inspection:** Counselors access assigned students, inspect academic levels, test scores, skills, and trigger AI student summaries.
- **Admin Platform Governance:** Administrators monitor user directories, approve/reject counselors, and track platform metrics.

38 - 61. DATABASE PERSISTENCE & FINAL SUMMARY

VisionPath utilizes 13 dedicated PostgreSQL databases (`visionpath_auth_db`, `visionpath_user`, `visionpath_student`, `visionpath_counselor`, `visionpath_admin`, `visionpath_assessment`, `visionpath_ai`, `visionpath_career`, `visionpath_skill`, `visionpath_study_plan`, `visionpath_resume`, `visionpath_notification`, `visionpath_file`).

Final Summary: VisionPath is a completed, production-ready AI career and education guidance platform built with Java 21, Spring Boot 3.5.5, Spring Cloud Gateway, PostgreSQL, React 19, and Google Gemini AI, delivering end-to-end guidance from deterministic academic assessments to automated ATS resume audits and actionable 4-week study plans.